ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Section A: objective questions (30) · Section B: arrays and structures, non-blocking timing and state machines, interrupts, motor and servo control, closed-loop (PID) control, communication and debugging (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Code in this paper is written for an Arduino-compatible microcontroller (C/C++). Minor syntax errors will not be penalised if the logic is correct.
5. Write neatly and legibly. Indent your code clearly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A variable that is changed inside an interrupt service routine should be declared as …

A const. B static only. C void. D volatile.

1.1.2 The function used to attach an interrupt to a pin on an Arduino is …

A pinMode() B attachInterrupt()
C interrupt() D digitalInterrupt()

1.1.3 In a PID controller, the term that reacts to the accumulated error over time is the …

A proportional. B derivative. C integral. D offset.

1.1.4 The Servo library function that moves a servo to 90° is …

A myServo.write(90); B myServo.read(90);
C analogWrite(90); D myServo.attach(90);

1.1.5 A program design in which the robot is always in one of a fixed number of modes and changes mode on events is a …

A linked list. B state machine.
C recursive function. D library.

1.1.6 The index of the last element of the array int a[8]; is …

A 8. B 9. C 0. D 7.

1.1.7 A struct in C/C++ is used to …

A group related variables of different types under one name. B repeat code.
C store only one value. D connect to Wi-Fi.

1.1.8 The millis() counter on an Arduino overflows (starts again at 0) after about …

A 1 hour. B 24 hours. C 49,7 days. D 1 year.

1.1.9 Too much proportional gain (K_p) in a motor position controller usually causes …

A a slower response only. B no movement.
C overshoot and oscillation. D lower power use.

1.1.10 The bitwise operator that shifts the bits of x one place to the left is …

A x << 1 B x >> 1 C x & 1 D x ^ 1

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Interrupt service routines should be short and should not use delay().

1.2.2 Using delay() is the best way to run several tasks at the same time.

1.2.3 A switch–case statement is often used to implement a state machine.

1.2.4 Arrays in C/C++ automatically check that an index is inside the array.

1.2.5 The derivative term of a PID controller reacts to how fast the error is changing.

1.2.6 Global variables can be accessed from any function in the program.

1.2.7 An enum gives meaningful names to a set of integer constants.

1.2.8 The expression (currentMillis - previousMillis >= interval) still works correctly when millis() overflows, if unsigned long variables are used.

1.2.9 A pointer stores the address of a variable.

1.2.10 Serial.read() returns the whole line of text that was sent.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Function that runs automatically when an interrupt occurs	A	debouncing
1.3.2	Difference between the desired value and the measured value	B	constrain (clamp)
1.3.3	The desired value a controller tries to reach	C	error
1.3.4	Code that does not stop the program while it waits	D	queue (FIFO buffer)
1.3.5	Averaging several readings to reduce noise	E	setpoint
1.3.6	A set of named integer constants, e.g. IDLE, DRIVE, TURN	F	enumeration (enum)
1.3.7	Value limited so that it stays within a range	G	non-blocking code
1.3.8	Removing false triggers from a mechanical switch	H	interrupt service routine (ISR)
1.3.9	Data structure in which the first value in is the first value out	I	flashing (uploading) firmware

1.3.10	Loading a new program into the board over a cable or wirelessly	J	filtering (smoothing)

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: ARRAYS, STRUCTURES AND NON-BLOCKING TIMING

2.1 A robot has five IR line sensors connected to pins A0 to A4.

2.1.1 Declare an array for the sensor pins and write a loop that reads all five sensors into a second array called values. (5)

2.1.2 Write code that finds the index of the sensor with the highest reading (the sensor over the line). (4)

2.2 Define a struct called Motor that stores the PWM pin, the direction pin and the current speed. Declare two Motor variables, left and right, with suitable pin numbers. (4)

2.3 Write non-blocking code that toggles a status LED on pin 13 every 250 ms while the rest of loop() continues to run. (6)

[19]

QUESTION 3: STATE MACHINES AND INTERRUPTS

3.1 A sorting robot has three states: WAIT (waiting for an object), MEASURE (reading its colour sensor for 1 second) and SORT (moving a servo to 30° for red or 150° for other colours, then back to 90°).

3.1.1 Draw a state diagram showing the states and the events that cause each transition. (5)

3.1.2 Write the loop() code for the state machine using an enum and a switch statement. Use the functions objectDetected(), readColour() (returns 'R' for red) and the servo object sorter. (10)

3.2 A wheel encoder gives 20 pulses per revolution on pin 2.

3.2.1 Write the code to count the encoder pulses using an interrupt. (5)

3.2.2 The wheel has a diameter of 65 mm. Calculate the distance travelled after 140 pulses. (3)

[23]

QUESTION 4: MOTOR CONTROL AND PID

4.1 Write a function setMotor(int speed) that accepts a speed from −255 to 255, sets the direction pin (pin 7) HIGH for forward and LOW for reverse, and writes the size of the speed to the PWM pin (pin 6). (6)

4.2 A line-following robot uses a proportional controller. The error is the position of the line relative to the centre (−2 to +2). The base speed is 150 and $K_p = 40$.

4.2.1 Write the formulas for the left and right motor speeds. (2)

4.2.2 Calculate the left and right speeds when the error is +1,5. (2)

4.2.3 Explain what adding the derivative term would improve. (2)

4.3 Explain the effect of increasing each of K_p, K_i and K_d on the robot's behaviour. (6)

[18]

QUESTION 5: COMMUNICATION AND DEBUGGING

5.1 A robot receives commands over Bluetooth serial as single characters: 'F' forward, 'B' back, 'L' left, 'R' right, 'S' stop. Write the code that reads and acts on the commands. Use the functions forward(), back(), left(), right() and stopAll(). (6)

5.2 Give FOUR techniques for debugging a robot program. (4)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER